

Risk factors for bleeding complications after percutaneous dilatational tracheostomy: a ten-year institutional analysis

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Summary

Bleeding complications after percutaneous dilatational tracheostomy (PDT) are infrequent but may have a tremendous impact on a patient's further clinical course. Therefore, it seems necessary to perform risk stratification for patients scheduled for PDT. We retrospectively reviewed the records of 1001 patients (46% male, mean age 68.1 years) undergoing PDT (using the Ciaglia Blue Rhino® technique with direct bronchoscopic guidance) in our cardiothoracic ICU between January 2003 and February 2013. Patients were stratified into two groups: patients suffering acute moderate, severe, or major bleeding (Group A) and patients who had no or only mild bleeding (Group B). In the majority of patients, no or only mild bleeding during PDT occurred (none: 425 [42.5%], mild: 488 [48.8%]). In 84 patients (8.4%), bleeding was classified as moderate. Three patients suffered from severe bleeding; only one major bleed with need for emergency surgery occurred. Patients in Group A had a significantly higher Simplified Acute Physiology Score on the day of PDT ($P=0.042$), higher prevalence of renal replacement therapy on the day of PDT ($P=0.026$), higher incidence of coagulopathy ($P=0.043$), lower platelet counts ($P=0.037$), lower fibrinogen levels ($P=0.012$), higher proportion of PDTs performed by residents ($P=0.034$) and higher difficulty grading of PDT ($P=0.001$). Using logistic regression analyses, difficult PDT, less experienced operator, Simplified Acute Physiology Score >40 and low fibrinogen levels were independent predictors of clinically significant bleeding after PDT.

Key Words: percutaneous dilatational tracheostomy, coagulopathy, bleeding

Tracheostomy is regularly considered as an airway access option in critically ill patients requiring prolonged mechanical ventilation to facilitate adequate airway management and ventilatory weaning. Since the introduction of guide-wire-assisted percutaneous dilatational tracheostomy (PDT) into routine clinical practice by Ciaglia et al, this technique or its modifications have been increasingly used in different patient populations^{1,2}. Although complication rates of PDT seem to be lower or comparable to open surgical tracheostomy, this approach is an invasive procedure with early complications including bleeding, infection, pneumothorax, technical failure and perioperative hypoxia due to tube obstruction or

accidental decannulation³. Bleeding can have a tremendous impact on further clinical course as bleeding or vascular injury account for the majority of fatal complications⁴.

Against this background, it seems necessary to perform risk stratification for patients scheduled for PDT, which is a common elective procedure. A number of potential risk factors for periprocedural complications during PDT have been stated, including coagulopathy, repeat PDT, less experienced operator, obesity and low puncture site^{5–7}. However, most studies included patients with percutaneous and open surgical tracheostomy, had a small sample size, and the definition and classification of complications varied significantly. Consequently, there is still controversy about the association of specific risk factors and complications during percutaneous tracheostomy. Therefore, it was the aim of our study to assess the risk factors for periprocedural bleeding of PDT in patients after cardiothoracic surgery.

Materials and methods

Patients

Between January 2003 and February 2013, data were prospectively collected on all consecutive patients undergoing PDT at our cardiothoracic ICU, in our institutional database. A retrospective analysis was then performed. The study was approved by the Institutional Review Board. All patients had previously granted permission for use of their medical records for research purposes. Within the observation period, all patients with expected prolonged mechanical

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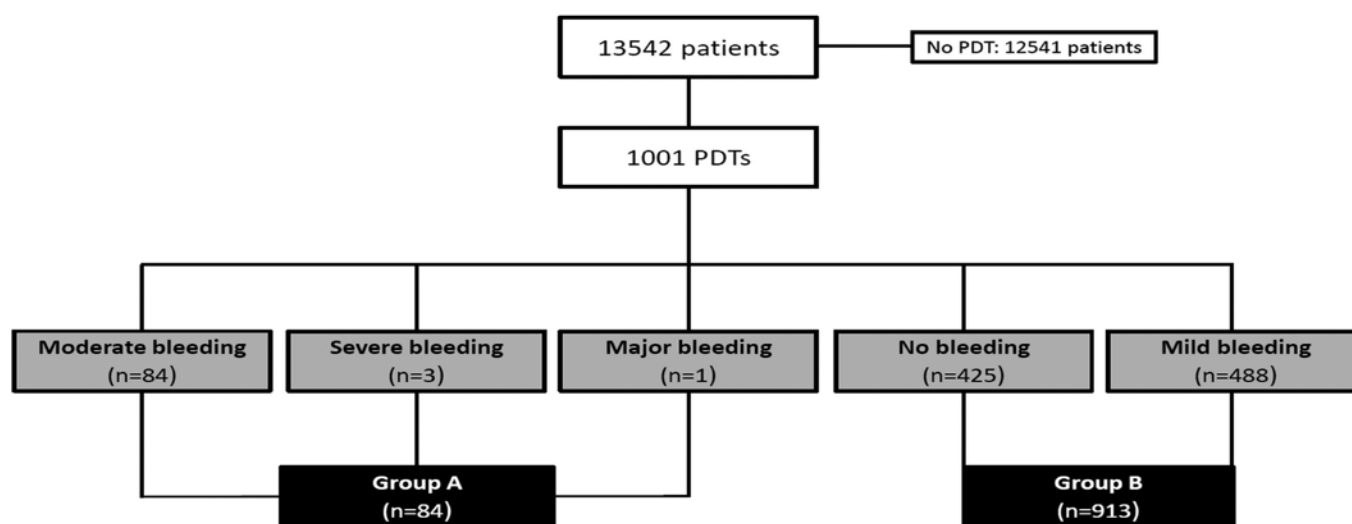


Figure 1: Description of the study population and classification of patients according to periprocedural bleeding complications.

ventilation were evaluated for PDT. Risk stratification of patients undergoing PDT was mainly based on medical history (e.g. difficult airway, coagulation abnormalities) as well as clinical assessment by the performing physician. Absolute contraindications to PDT in our institution included unstable fractures of the cervical spine and severe local infection of the anterior neck. In these cases, patients were primarily assigned to surgical tracheostomy. Other conditions, such as difficult anatomy (e.g. morbid obesity, short thick neck, reduced neck extension, excessive goitre and tracheal deviation), elevated intracranial pressure, haemodynamic instability and previous radiotherapy to the neck were considered as relative contraindications. To identify risk factors for periprocedural bleeding, patients were divided into patients suffering acute moderate, severe, or major bleeding (Group A) and patients who had no or only mild bleeding (Group B). Patients were considered to have a coagulopathy if one or more of the following criteria were met: 1) platelets $<50 \times 10^9/l$, 2) prothrombin time (PT) $>50\%$ of the normal reference value, and/or 3) activated partial thromboplastin time (aPTT) >50 seconds. Perioperative risk evaluation was based on two critical care scoring systems: the Simplified Acute Physiology Score (SAPS) and the Therapeutic Intervention Scoring System-28⁸. In addition, the European System for Cardiac Operation Risk Evaluation (EuroSCORE) II was calculated for every patient⁹. The EuroSCORE is the most commonly used tool to assess operative mortality risk in cardiac surgery. The first EuroSCORE, initially based on an additive system derived by a logistic regression model, was developed on data collected in 1995 and reported in 1999 to predict in-hospital or 30-day mortality. Recently,

in order to improve the poor calibration and to optimise EuroSCORE usefulness, an updated version of this model has been proposed for the assessment of cardiac surgical risk (EuroSCORE II).

Performance of PDT

The decision to perform tracheostomy was made by the executive consultant or the head of the department. The indication for tracheostomy was respiratory failure attributed to a variety of underlying medical conditions, most commonly exacerbation of chronic lung disease, muscle weakness, neurological deterioration, pulmonary infection or acute lung injury. The Ciaglia Blue Rhino[®] technique, with direct bronchoscopic guidance by the Tracoe Medical[™] system (experc Dilatation Set, Tracoe Medical GmbH, Nieder-Olm, Germany) was used in all patients. All procedures were performed at the bedside in the ICU by an experienced consultant in cardiothoracic surgery or anaesthesiology with specialisation in critical care (at least 50 PDTs performed) or a senior ICU trainee under consultant supervision. Two physicians and at least one nurse were always present; the cardiothoracic surgery team provided surgical cover. One physician provided video bronchoscopic guidance and the other performed the procedure. Informed consent was obtained prior to intervention from the patient's next of kin.

Patients were sedated and given analgesia as needed. Commonly used sedatives were remifentanyl, propofol and midazolam. Neuromuscular blockade was achieved in most cases with rocuronium. The fraction of inspired oxygen was increased to 1.0 and ventilator settings were kept unchanged; apart from patients on pressure support ventilation, in

whom a controlled mode of ventilation was initiated for the period of the intervention and maintained until reversal of paralysis. Continuous monitoring of heart rate, peripheral transcutaneous oxygen saturation and invasive arterial blood pressure was performed in all patients. Resuscitation and difficult airway equipment was present at the bedside.

Patients were placed in the supine position with their necks extended and were prepared and draped in a sterile manner. Routine pre-PDT ultrasound screening was not performed during the whole observation period but was established in 2013. Flexible bronchoscopic visualisation was routinely used to facilitate insertion in all patients. A bronchoscope was advanced to the end of the endotracheal tube (ETT). After deflating the cuff, the ETT was slowly withdrawn until the bronchoscope light illuminated the planned tracheostomy site or digital pressure of the trachea over the site could be seen through the bronchoscope.

A 16-gauge introducer needle with introducer sheath was advanced into the trachea in the midline between the first and second or the second and third tracheal cartilage rings. The needle was removed and a J-wire was advanced inferiorly through the introducer sheath, then the introducer sheath was removed. A small skin incision (approximately 1.0 to 1.5 cm) was made with a scalpel at the entry site of the wire. A 14F introducer dilator was then advanced. Following this, one large hydrophilic dilator with a guiding catheter was advanced over the wire until the stoma was appropriately dilated for passage of the cannula and the dilator was removed. A tracheostomy cannula—loaded onto a guiding catheter—was advanced over the J-wire into the trachea followed by balloon insufflation. The guiding catheter and J-wire were removed, leaving the tracheostomy cannula in place. Ventilation was then switched from the ETT to the tracheostomy tube. Regions of the trachea superior and inferior to the cannula were examined, blood was aspirated if necessary, and the ETT was removed. The tracheostomy tube was secured with neck ties. Chest X-ray was routinely performed after PDT to rule out a pneumothorax. Coagulopathy, if present before percutaneous tracheostomy, was not routinely corrected.

Statistical analysis

Statistical analyses were performed with SPSS Statistics 19 (IBM, Chicago, IL, USA). Continuous data were expressed as mean $\pm$ standard deviation; categorical data were expressed as percentages. Comparisons between the two groups were carried out using unpaired student's t-test for normally, or the Mann–Whitney Rank Sum Test for non-normally, distributed data. Multiple groups were compared with analysis of variance. Univariate analysis was performed on the quantitative variables using student's t-test or the Mann–Whitney U test, and on the qualitative variables using the chi-square test or Fisher's exact test. All variables showing a *P*-value of less than 0.1 between the two groups using student's t-test, Mann–Whitney U test, chi-square test

or Fisher's exact test were selected for univariate analysis. In addition, previously published risk factors for bleeding complications such as obesity were included, even if they did not meet the criteria mentioned above. Variables with *P* < 0.1 in the univariate analysis were included in a logistic regression model to determine independent factors associated with bleeding complications. Model calibration was tested using the Hosmer–Lemeshow test. Statistical significance was assumed for a *P*-value < 0.05.

Assessment of complications

Complications related to PDT such as death, hypoxaemia, pneumothorax, bleeding, tracheal injuries (posterior wall perforation, fractures of tracheal rings), cannula dislocation, premature decannulation and stomal infections were documented. Acute bleeding was defined as endo- or extrabronchial bleeding occurring during tracheostomy (between skin incision and insertion of the tracheostomy tube). Acute bleeds were subclassified as mild (extrabronchial: 1 to 5 ml, endobronchial: 1 to 2 blood-covered tracheal rings or presence of some blood on the posterior tracheal wall not requiring intratracheal suction), moderate (extrabronchial: 5 to 20 ml, endobronchial: PDT site blood-covered or segmental bronchus obstructed by blood), severe (extrabronchial: 20 to 50 ml, endobronchial: mainstem bronchus obstructed by blood) and major (>50 ml and/or surgical intervention/transfusion required). The volume of periprocedural extrabronchial blood loss was calculated by measuring the difference in weight of the gauzes before and after the procedure. Insertion of the tracheostomy tube was graded as easy, moderately difficult, very difficult or impossible based on subjective assessment by the performing physician rather than objective criteria. Obesity was defined as a body mass index >30 kg/m²; low PDT puncture was defined as puncture below the fourth tracheal ring. Patients were followed up until discharge from the hospital.

Results

During the observation period, a total of 13,542 patients were admitted to our cardiothoracic ICU. Out of these, 1012 patients (7.5%) were scheduled for tracheostomy (see Figure 1). Excluding patients primarily assigned for open surgical tracheostomy due to contraindications for PDT (*n*=11), 1001 patients underwent PDT with a success rate of 99.8% (999/1001). No patients were lost to follow-up.

One patient had major bleeding due to a periprocedural laceration of the brachiocephalic trunk with need for emergency surgery; three patients had severe bleeding; and 84 (8.4%) were classed as having moderate bleeding (Group A, *n*=88). Bleeding was classed as nil in 425 patients (42.5%) and mild in 488 (48.8%), thus there were 913 patients in Group B (Figure 1).

There were no deaths associated with PDT, and no patient

suffered a major complication, such as cardiac arrest, hypoxaemia, accidental extubation or pneumothorax. Other early complications were observed infrequently: misplacement in one patient (0.1%), posterior wall injury in two patients (0.2%), and hypotension in three patients (0.3%). Fracture of the tracheal ring occurred in 24/84 (28.6%) and 295/913 (32.3%) without any effect on further clinical course. Late complications included purulent stomal infection in three cases (0.3%), mediastinitis in 34 cases (3.4%) and development of granuloma in two patients (0.2%).

Table 1 shows the main characteristics of patients with bleeding complications compared to those without. Whereas baseline demographics including age, gender, body mass index, EuroSCOREII, SAPS and Therapeutic Intervention Scoring System-28 on the day of admission to ICU as well as

type of surgery did not differ between the two groups; SAPS on the day of PDT was significantly higher in Group A (47.0 ± 13.1 versus 32.9 ± 11.2 , $P=0.042$). Renal replacement therapy (RRT) on the day of PDT was more frequently performed in patients with bleeding complications (53 [60.2%] versus 439 [48.1%], $P=0.026$). ICU stay (25.8 ± 26.6 versus 24.8 ± 22.4 days), as well as hospital mortality (29 [32.9%] versus 284 [31.1%]) were comparable between the two study groups.

Whereas activated clotting time, aPTT, and PT did not differ between the two study groups, mean platelet count and mean fibrinogen levels were significantly lower in the bleeding group (platelet count: 91.6 ± 59.2 versus $111.5 \pm 79.8 \times 10^9/l$, $P=0.037$; fibrinogen level: 10.9 ± 4.7 versus $13.3 \pm 7.6 \mu\text{mol/l}$, $P=0.012$, Table 2). Accordingly, proportions of patients with platelets $<80 \times 10^9/l$ (51 [57.9%] versus 378

Table 1
Patient characteristics comparing patients with and without moderate to major bleeding complications

	Group A Acute Bleeding (n=88)	Group B No Bleeding (n=913)	P-value
Age (years)	70.5±10.6	70.5±11.9	n.s.
BMI (kg/m ²)	26.8±4.6	27.1±6.2	n.s.
Obesity (n, %)	14 (15.9)	190 (20.4)	n.s.
Male gender (n, %)	57 (64.7)	596 (65.3)	n.s.
Time from intubation to PDT (days)	3.7±3.2	3.8±3.3	n.s.
Reintubation (n,%)	36 (40.9)	376 (41.2)	n.s.
RRT (n, %)	53 (60.2)	439 (48.1)	0.026
SAPS, day of surgery	37.3±8.1	33.1±12.6	n.s.
SAPS, day of PDT	47.0±13.1	32.9±11.2	0.042
SAPS >40 (n, %)	60 (68.2)	230 (25.2)	<0.001
TISS-28, day of surgery	23.0±5.2	23.0±12.1	n.s.
TISS-28, day of PDT	23.0±5.2	21.9±8.8	n.s.
EuroSCORE II (%)	10.1±11.1	10.9±11.6	n.s.
<i>Surgery (n, %)</i>			n.s.
CABG	36 (40.9)	343 (37.6)	n.s.
AVS	10 (11.4)	108 (11.8)	n.s.
MVS	5 (5.7)	44 (4.8)	n.s.
Combined surgery	27 (30.7)	266 (29.1)	n.s.
Aortic surgery	5 (5.7)	89 (9.7)	n.s.
Thoracic transplant	1 (1.1)	20 (2.2)	n.s.
Others	4 (4.5)	43 (4.7)	n.s.
ICU stay (days)	25.8±26.6	24.8±22.4	n.s.
Hospital mortality (n, %)	29 (32.9)	284 (31.1)	n.s.

BMI=body mass index, PDT=percutaneous dilatational tracheostomy, RRT=renal replacement therapy, SAPS=Simplified Acute Physiology Score, TISS-28=Therapeutic Intervention Scoring System 28, EuroSCORE=European System for Cardiac Operation Risk Evaluation, CABG=coronary artery bypass grafting, AVS=aortic valve surgery, MVS=mitral valve surgery, n.s.=not significant.

[41.4%], $P=0.007$) and patients with low fibrinogen levels <7.4 $\mu\text{mol/l}$ (27 [30.7%] versus 171 [18.7%], $P=0.022$) was higher in the group of patients with bleeding complications as well as the proportion of patients with coagulopathy (48 [54.5%] versus 393 [43.0%], $P=0.043$). There was no significant difference in the prevalence of antiplatelet therapy (aspirin or clopidogrel as monotherapy or both as dual therapy) between patients with bleeding during tracheostomy and those without.

Detailed technical information about PDT comparing patients with and without bleeding complications are given in Table 3. Most PDT tube insertions were graded as easy or moderately difficult in both groups. However, grading of PDT as moderately or very difficult was more frequently observed in patients with bleeding complications (31 [35.2%] versus 141 [15.4%], $P=0.001$). The majority of all tubes were placed between the 1st and 2nd, or 2nd and 3rd ring, without any group difference. In 36 cases, patients had undergone previous PDT with equal distribution between the two groups. In Group A, statistically more PDTs were performed by residents than in Group B (72 [81.8%] versus 632 [69.2%], $P=0.034$).

The results of univariate and multivariate logistic regression analyses are presented in Table 4. In univariate analyses, difficult PDT (odds ratio [OR] 2.98, 95% confidence interval [CI] 1.86 to 4.78, $P<0.001$), resident-status of the operator (OR 2.00, 95% CI 1.4 to 3.50, $P=0.015$), low fibrinogen

<7.4 $\mu\text{mol/l}$ (OR 1.92, 95% CI 1.19 to 3.11, $P=0.008$), low platelets $<80 \times 10^9 /\text{l}$ (OR 1.95, 95% CI 1.25 to 3.04, $P=0.003$), coagulopathy (OR 1.56, 95% CI 1.01 to 2.43, $P=0.046$), need for RRT on the day of PDT (OR 1.64, 95% CI 1.05 to 2.55, $P=0.0308$), and SAPS >40 (OR 6.87, 95% CI 4.24 to 10.54, $P<0.001$) were predictors of moderate to major bleeding complications. In contrast, obesity, repeat PDT and low PDT puncture site did not affect incidence of bleeding complications. After adjusting for all significant factors in univariate analyses, difficult PDT (OR 2.01, 95% CI 1.27 to 3.20, $P=0.003$), resident-status of the operator (OR 2.10, 95% CI 1.12 to 3.97, $P=0.021$), SAPS >40 (OR 6.40, 95% CI 3.79 to 10.82, $P<0.001$) and low fibrinogen (OR 1.85, 95% CI 1.05 to 3.27, $P=0.033$) remained independent predictors of moderate to major bleeding complications after PDT.

Discussion

The present study demonstrated that low fibrinogen levels as well as difficult PDT, less experienced operator and SAPS >40 are independent predictors of moderate to major bleeding complications after PDT.

The overall incidence of bleeding complications—including moderate as well as severe bleeding events—was 8.8 % in our series, being somewhat higher than in recent reports^{10,11}. However, it has to be mentioned that it is difficult to compare different studies about bleeding complications

Table 2

Coagulative status of patients with and without moderate to major bleeding complications

	Group A Acute Bleeding (n=88)	Group B No Bleeding (n=913)	P-value
ACT (s)	126.0 $\pm$ 21.9	124.3 $\pm$ 22.8	n.s.
Platelets ($\times 10^9 /\text{l}$)	91.6 $\pm$ 59.2	111.5 $\pm$ 79.8	0.037
Platelets $<80 \times 10^9 /\text{l}$ (n, %)	51 (57.9)	378 (41.4)	0.007
aPTT (s)	48.0 $\pm$ 18.9	45.8 $\pm$ 16.1	n.s.
aPTT >50 s (n, %)	28 (31.8)	239 (26.2)	n.s.
PT (% normal reference)	70.1 $\pm$ 22.6	78.8 $\pm$ 59.5	n.s.
PT $>50\%$ normal reference (n, %)	16 (18.2)	160 (17.5)	n.s.
Fibrinogen (mean, $\mu\text{mol/l}$)	10.9 $\pm$ 4.7	13.3 $\pm$ 7.6	0.012
Fibrinogen <7.4 $\mu\text{mol/l}$ (n, %)	27 (30.7)	171 (18.7)	0.022
Mono antiplatelet therapy (n, %)	52 (59.0)	526 (57.6)	n.s.
Dual antiplatelet therapy (n, %)	6 (6.8)	44 (4.8)	n.s.
Coagulopathy (n, %) *	48 (54.5)	393 (43.0)	0.043

ACT=activated clotting time, aPTT=activated partial thromboplastin time, PT=prothrombin time, n.s.=not significant.

*one or more of the following criteria were met: 1) platelets $<50 \times 10^9 /\text{l}$, 2) prothrombin time (PT) $>50\%$ of the normal reference value, and/or 3) activated partial thromboplastin time (aPTT) >50 seconds.

Table 3

Technical details of PDT comparing patients with and without moderate to major bleeding complications

	Group A Acute Bleeding (n=88)	Group B No Bleeding (n=913)	P-value
<i>Puncture site between (n, %)</i>			
1st/2nd ring or 2nd/3rd ring	69 (78.4)	786 (86.1)	n.s.
3rd/4th ring or below	15 (17.9)	110 (12.4)	n.s.
<i>Difficulty of insertion (n, %)</i>			
Easy	57 (64.8)	785 (85.9)	0.001
Moderately difficult	27 (30.7)	108 (11.8)	0.001
Very difficult	2 (2.3)	33 (3.6)	n.s.
Impossible	2 (2.3)	0 (0)	n.s.
Other than easy	31 (35.2)	141 (15.4)	0.001
<i>Tracheostomy tube used (n, %)</i>			
8 mm ID	32 (36.4)	346 (37.9)	n.s.
9 mm ID	56 (63.6)	567 (62.1)	n.s.
Success rate	86 (97.7)	931 (100)	n.s.
Performed by resident	72 (81.8)	632 (69.2)	0.034
Repeat PDT	3 (3.4)	33 (3.6)	n.s.

PDT=percutaneous dilatational tracheostomy, ID=internal diameter, n.s.=not significant.

Table 4
Univariate and multivariate analysis of risk factors for acute moderate to major bleeding

	Univariate analysis			Multivariate analysis		
	OR	95% CI	P-value	OR	95% CI	P-value
Difficult PDT	2.98	1.86–4.78	<0.001	2.01	1.27–3.20	0.003
Resident	2.00	1.43–3.502	0.015	2.11	1.12–3.97	0.021
Obesity*	0.72	0.39–1.33	0.277	-	-	-
Re-PDT	0.94	0.28–3.13	0.921	-	-	-
Fibrinogen <7.4 µmol/l	1.92	1.19–3.11	0.008	1.85	1.05–3.27	0.033
Platelets <80 × 10 ⁹ /l	1.95	1.25–3.04	0.003	1.52	0.88–2.61	0.133
Coagulopathy†	1.56	1.01–2.43	0.046	1.45	0.84–2.48	0.180
SAPS >40	6.87	4.24–10.54	<0.001	6.40	3.79–10.82	<0.001
Low puncture‡	1.50	0.83–2.71	0.178	-	-	-
RRT	1.64	1.05–2.55	0.0308	1.42	0.71–2.82	0.144

*Body mass index >30 kg/m². † prothrombin time >50% of the normal reference value, activated partial thromboplastin time >50 seconds, and/or platelet count <50 × 10⁹ /l. ‡ PDT below 3rd tracheal ring. PDT=percutaneous tracheostomy, SAPS=Simplified Acute Physiology Score, RRT=renal replacement therapy.

as the definition and classification of bleeding is not consistent throughout the studies and some did not include endobronchial bleeding at all. As our definition of bleeding includes both endobronchial and extrabronchial bleeding and the grading is very strict, this may be at least partly responsible for the higher overall incidence of bleeding events in our cohort.

There is little data in the literature about the impact of coagulopathy/thrombocytopenia on bleeding events after PDT. Beiderlinden investigated the effect of the preoperative coagulation status on the incidence of acute and chronic bleeding in 415 consecutive patients¹¹. Acute bleeding was not related to coagulopathy in general, but prolonged aPTT, a low platelet count and abnormalities of two or more of aPTT, platelets and PT significantly increased the risk for chronic bleeding. One limitation of this study is the fact that different techniques of PDT were used during the observation period. In another retrospective study, the number of bleeding complications related to PDT was minimal, despite elevated aPTT or International Normalized Ratio (INR) or thrombocytopenia¹⁰. A prospective cohort study with 60 consecutive patients in need of tracheostomy insertion showed that PDT in coagulopathic patients with liver disease or following liver transplantation for acute liver failure is safe¹². Clinically relevant bleeding complications were infrequent, even in patients with refractory coagulopathy¹².

Although patients after cardiac surgery represent a high risk population as most of them require adequate anticoagulation, especially after valve replacement and with any kind of assist device, data about the association of bleeding complications during PDT and coagulation disorders after cardiac surgery are rare. Takahashi reported no bleeding complications in 149 patients undergoing PDT, 75 of whom

(49%) were coagulopathic¹³. Another retrospective study with 213 patients (85 with coagulopathy) could not detect any impact of impaired haemostasis on periprocedural bleeding complications¹⁴.

In our study, coagulopathy was more frequently observed in patients with moderate to major bleeding after PDT, and coagulopathy as well as low platelets and low fibrinogen levels represented a risk factor for bleeding events in univariate analysis. In multivariate analyses, only low fibrinogen levels remained a significant predictor. To our knowledge, this is the first study describing an association of fibrinogen levels and bleeding after PDT. All studies including ours did not include fibrinogen in their primary definition of coagulopathy but used platelets, PT or INR and aPTT as criteria for a coagulation disorder. In addition, no other study reported fibrinogen levels at all. A fibrinogen level of 2.9 µmol/l is traditionally considered to be the threshold plasma concentration required for adequate haemostasis. Recently published data have shown an increased peri- and postoperative bleeding risk in different patient populations with low plasma fibrinogen levels¹⁵. A recent study showed that a low fibrinogen level is independently associated with injury severity and replacement with cryoprecipitate improved survival in trauma patients¹⁶. Based on our results, low fibrinogen levels are an independent predictor for acute bleeding complications during PDT.

Thrombocytopenia is a common finding in critically ill patients, often associated with a higher incidence of bleeding, prolonged ICU stay and increased mortality¹⁷. This applies, in particular, for patients undergoing cardiac surgery, with cardiopulmonary bypass resulting in an initial drop in the platelet count of approximately 50% on the initiation of bypass, and a nadir of platelet count on the second or third day after

surgery¹⁷. There is little data about the association of thrombocytopenia and bleeding complications after PDT. Beiderlinden et al found thrombocytopenia to be the strongest single risk factor of chronic bleeding and recommended platelet transfusion when the platelet count is below $50 \times 10^9 / l$ ¹¹. Kluge et al described a low complication rate in patients with severe thrombocytopenia, if platelets were administered beforehand¹⁸. In our study, mean platelet count was significantly lower in the group with bleeding complications and low platelet count was a risk factor for bleeding events in univariate, but not in multivariate, analysis.

In addition, no association between the intake of platelet inhibitors and bleeding complications was observed, supporting the hypothesis that tracheostomy during (dual) antiplatelet therapy can be safely performed and that interruption of antiplatelet therapy is not required¹⁹.

Repeat PDT is regularly cited as a relative contraindication due to distortion of the normal anatomy with difficulty in identifying landmarks and risk of excessive bleeding. In contrast, there are no published data supporting this hypothesis and several case reports describe the successful performance of PDT in patients with previous tracheostomy²⁰. Based on our experience, repeat PDT is a safe procedure in experienced hands and under controlled conditions and should not be generally considered a contraindication. However, the only patient suffering from a near-fatal complication in our series underwent repeat PDT. Therefore, special attention should be paid to the anatomical situation with an increased risk of vascular complications.

Incidence of complications after invasive procedures does not only depend on the inherent problems of the technique, but also on the experience of the performing physician. A learning curve for percutaneous tracheostomy is described in the literature with a decreased rate of complications and procedure time with increasing experience being most prominent in the first 20 patients²¹. Congruently, Petros observed a threefold reduction in the complication rate after two years of practising²². Our results are consistent with these reports, identifying a less experienced physician performing PDT as an independent risk factor for bleeding complications. However, the overall low rate of complications and the high number of PDTs performed by residents rather than by ICU staff physicians indicate that PDT can be performed safely and can be considered as a routine procedure in the care of ICU patients in daily practice by physicians during their training.

Since the introduction of PDT into clinical practice, obesity has been considered as a relative contraindication based on the assumption that anatomical landmarks might be more difficult to identify due to larger and thicker necks in obese patients. However, studies addressing the safety of PDT in obese patients are contrary. Mansharamani reported PDT successfully performed on 13 consecutive obese

patients with a body mass index $>27 \text{ kg/m}^2$ with minimal complications²³. Romero et al compared 25 obese patients (body mass index $>30 \text{ kg/m}^2$) with 80 non-obese patients undergoing PDT with no significant difference between the two groups in terms of complications²⁴. These findings are supported by recently published studies²⁵. In contrast, Byhahn suggested that PDT in obese patients is almost five times more likely to result in serious adverse events⁵. However, observed serious complications included posterior tracheal wall injury, airway loss and cardiac arrest but not bleeding. The largest series with 1175 PDTs identified obesity and the use of large endotracheal tubes (size $>7.5 \text{ mm ID}$) as major risk factors for complications such as development of airway stenosis, but not for bleeding events²⁶. In our series, no association between obesity and bleeding complications could be observed, confirming the safety of PDT in these patients as regards bleeding.

Recent reviews demonstrated that in a significant number of patients suffering from fatal bleeding complications after PDT, the tracheostomy tube was positioned too low (in the majority of patients below the sixth tracheal ring)⁶. In our study, low puncture site was not a risk factor for bleeding complications. However, all of our PDTs were performed under fiberoptic visualisation during the whole procedure. Therefore, we could not observe puncture sites as low as described in the cited review and cannot draw a clear conclusion about the association of a very low puncture site and bleeding complications.

In our study, grading of PDT as moderately or very difficult was an independent predictor for bleeding complications. However, this finding has to be interpreted with caution; grading of PDT difficulty was based on the subjective assessment of the performing physician rather than objective criteria. Therefore, it cannot be ruled out that bleeding complications observed during PDT served as a criterion for difficult tracheostomy.

In addition, severity of illness assessed by the SAPS was a strong independent predictor for bleeding complications. The SAPS II is a severity score and mortality estimation tool developed from a large sample of medical and surgical patients in North America and Europe²⁷. The SAPS II score is made up of 12 physiological variables and three disease-related variables but no coagulation parameters. Coagulation abnormalities are commonly found in critically ill patients, of which some are measurable, such as thrombocytopenia, prolonged global coagulation times or reduced levels of coagulation inhibitors²⁸. However, routine coagulation tests might miss clinically significant coagulation defects that can contribute to bleeding. The most important coagulation defects after cardiac surgery that may remain undetected with routine coagulation tests are platelet dysfunction and hyperfibrinolysis²⁹. Therefore, it might be hypothesised that the degree of these 'non-measurable' coagulation

disorders might be correlated to the severity of illness. Studies with septic patients did show that severity of haemostatic imbalance was significantly correlated to the severity of illness assessed by the SAPS³⁰.

RRT on the day of PDT was identified as a risk factor in univariate, but not multivariate, analysis. There are some possible explanations for this finding. For example, RRT can be considered as a surrogate marker for a high severity of illness and be related to coagulation disorders by this pathway. In addition, RRT requires some kind of anticoagulation, regionally with citrate or systemically with heparin; in the latter case, with a consequent coagulation disorder. However, most of our patients with a need for RRT early after cardiac surgery are treated with local citrate-based anticoagulation for the many proposed advantages, such as prolonged filter life while decreasing RRT downtime, transfusion rates, and supplementation needs for antithrombin III and platelets.

In the current literature, prevalence of PDT after cardiac surgery ranges between 2.2% and 6%^{31,32}. Due to our experience with this technique and our low complication rate, we use a very liberal approach to early tracheostomy. Therefore, in our cohort, 1001 patients (7.4%) underwent percutaneous tracheostomy for prolonged mechanical ventilation. Although mean time from intubation to PDT might appear very short, with less than four days as the mean, it has to be kept in mind that a significant proportion had undergone reintubation, and mean time from surgery to PDT was approximately six days. Against the background of missing evidence-based recommendations, there is no clear cut-off for optimal timing of PDT and in most cases we consider tracheostomy after seven to ten days of mechanical ventilation. However, in selected cases with specific conditions, PDT is considered within in the first postoperative days. This applies in particular for patients with major intraoperative complications, preoperative mechanical ventilation or critical status, and severe perioperative neurological complications. In these specific patient groups, we believe to be able to predict the need for long-term mechanical ventilation at a very early time. This practice might not be routine in all centres but others also consider very early tracheostomy, such as in severely neurologically compromised patients, as it appears easier to predict need for long-term ventilatory support compared to other patients populations and might offer a mortality advantage. The randomised pilot trial *Stroke-related Early Tracheostomy versus Prolonged Orotracheal Intubation in Neurocritical Care Trial (SETPPOINT)* showed that use of sedatives, ICU mortality and six-month mortality were lower in the early PDT group than in the standard group³³. Others have also identified subgroups of patients benefiting from very early PDT. Rumbak et al showed a benefit in medical ICU patients with Acute Physiology and Chronic Health scores >25 when tracheostomy was performed very early (within 48 hours)

compared to late tracheostomy (14 days)³⁴.

Limitations

This is a retrospective, single-centre study with design-specific limitations. One major limitation of the present study is the lack of randomisation of treatment versus no treatment of coagulation disorders. Therefore, we cannot draw any conclusions about a potential benefit of pre-PDT correction of coagulation disorders, especially about replacement of fibrinogen. A prospective randomised trial evaluating the effect of preprocedural administration of fresh frozen plasma in patients with a prolonged INR undergoing placement of a central venous catheter, percutaneous tracheostomy, chest tube or abscess drainage could not detect any difference in bleeding complications³⁵. These findings are confirmed by another randomised trial demonstrating that correction of subclinical coagulation disorders by transfusion of fresh frozen plasma and/or platelets does not affect bleeding, either during or after PDT³⁶.

As all PDTs in this series were performed under bronchoscopic guidance, our results might not be applicable to institutions not using routine bronchoscopy. Although there is a lack of evidence in the literature, blind PDT without direct visualisation of the trachea might be associated with an increased risk of uncentred tracheal positioning, paratracheal insertion and posterior tracheal wall laceration or perforation, with subsequent potential for increased incidence of complications based on our experience.

Previous studies have emphasised the usefulness of ultrasound in addition to clinical examination to detect cases of abnormal pre-tracheal vascular anatomy to avoid bleeding complications during and after PDT. The only randomised controlled trial assigning a total of 50 patients to have tracheal puncture carried out using either traditional anatomical landmarks or real-time ultrasound guidance demonstrated that ultrasound guidance significantly improved the rate of first-pass puncture and puncture accuracy³⁷. There was a trend towards fewer procedural complications without statistical significance. As we only established routine pre-PDT ultrasound screening of the ventral neck with special attention to the thyroid gland and subcutaneous vessels recently, we cannot make any statements about the impact of pre-PDT ultrasound on the incidence of bleeding complications.

As mentioned above, platelet dysfunction and hyperfibrinolysis can be missed by routine coagulation tests. Therefore, we cannot draw any conclusions about the association of these non-measurable coagulation disorders and bleeding complications.

Conclusion

Periprocedural bleeding complications during PDT are

rare. However, low fibrinogen levels as well as difficult PDT, less experienced operator and SAPS >40 are associated with an increased risk for bleeding complications. Therefore, preprocedural risk evaluation for bleeding should include these factors and further studies are necessary to prove if modification of adaptable risk factors, e.g. replacement of fibrinogen prior to PDT, is able to reduce incidence of bleeding complications.

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